

Program DutchSemCor Workshop

Thursday 25 October 2012

Stoa room (2nd floor), VU University Amsterdam, De Boelelaan 1105, 1081 HV Amsterdam

13.00 – 13.20 Welcome (“Galerij”)

13.30 – 15.00 Methodology and tools (“Stoa”)

---Opening by Piek Vossen

Methodology, approaches and tools (Attila Görög)

Sense-tagged corpora should ideally represent all the senses of words (including rare senses), the variety of contexts, and provide information on sense distribution. These three requirements are often contradictory in practice. Our approach of trying to meet these three requirements in DutchSemCor consisted of three phases: 1.) we manually created a lexical-sample corpus with double annotations that met the first requirement; 2.) we semi-automatically extended this corpus through Active Learning after training a supervised WSD system using the data from the first phase and letting the WSD system select more examples to be validated by annotators; 3.) finally, we applied bootstrapping to automatically annotate the whole SoNaR corpus and to add more examples that better represent the context variety and the sense-distribution of the selected lemmas.

Manual annotation user experiences (Marlisa Hommel & Chantal van Son)

The initial training data has been created by student assistants of a linguistic background. Each token used for training was tagged by two annotators. In order to reach an inter-annotator agreement (IAA) of 80% or higher regular discussions took place to clarify problematic senses and tokens. In this presentation two annotators will share their experience with manual annotation of the DutchSemCor data and the role of domain labels.

Manual annotation statistics (Attila Görög)

In this short presentation we will provide some figures on the different subcorpora annotated manually: 1st cycle (human annotations made independent of the WSD system); 2nd cycle (human annotations made through active learning using the TiMBL system); All-words (human annotations made in the all-words corpus for evaluating the WSD systems); Manual annotations (containing statistics for all human annotations).

Semantic Annotation Tool (Fons Laan)

We will give a short demo on our Semantic Annotation Tool. The tool provided human annotators with an ergonomic and easy to use web-based environment for semantic annotation. The tool gives access to the Cornetto database and to text fragments from the reference corpus. Based on different types of information (definitions, examples, grammatical and semantic information), human annotators are required to link corpus examples to Cornetto-senses.

15.00 – 15.30 Coffee break (“Galerij”)

15.30 – 17.00 WSD systems and results (“Stoa”)

Corpora and the FoLiA format (Maarten van Gompel)

At the beginning of the project, we indexed and made available the first two releases of the Dutch Corpus, SoNaR. As a following step, a domain classification was conducted at the paragraph level. Also a corpus annotation format was developed by the UvT named FoLiA.

Named Entity Recognition using DutchSemCor data (Wouter Weerkamp)

Named Entity recognition and Wikification were carried out on the whole corpus but independent of the Cornetto database. Each Named Entity received a link to the corresponding Wikipedia page if present. Besides representing a separate semantic annotation, the Named Entities can also be used as features for WSD.

WSD, Domain classification and results (R. Izquierdo)

The WSD systems (TiMBL-WSD, SVM-BC-WSD and UKB-WSD) built from the final sense-annotated corpus were tested using an independent all-words corpus. UKB-WSD was trained using a large number of semantic relations derived from the manually-tagged data. The different WSD systems and their evaluation will be described.

Word Matrix (P. Vossen)

We will describe the deliverable Target Words Matrix which includes statistics for all target words and their annotations in the DutchSemCor-project. There are two files (.csv and .txt) for the three word categories (nouns, verbs and adjectives). Information on the number of instances, accuracy of the three WSD systems in the cross-fold evaluation as well as scores for the all-words evaluation can be found per lemma.

Closing by Antal van den Bosch---

17.00 – 18.00 Drinks (“Galerij”)